

CPE342 Syllabus

Couse Name: Machine Learning

Course Credits: 3 (2-2-6)

Semester/Year: 1/2025

Class Time: Friday 13.30 – 17.30, Room CB2506

Resource: LEB2 platform

Instructor: Dr. Boonyarit Changaival, boonyarit.chang@kmutt.ac.th

Office Hours: By appointment (via email) only.

Teaching Assistants: TBA

Course Description:

Theoretical basis of machine learning through examples and hands-on coding exercises. Machine learning techniques for supervised learning, unsupervised learning, and reinforcement learning. Statistical learning, Model training, Kernel machines, Regression and Generalized linear models, Tree and Ensemble models, Dimensionality reduction, Clustering, Neural network, Deep learning and applications

Course Learning Outcomes (CLOs):

After completing this course, the student should be able to

CLO1: Demonstrate mastery in concepts and details of supervised learning algorithms.

CLO2: Demonstrate mastery in concepts and details of unsupervised and reinforcement learning algorithms.

CLO3: Implement machine learning models and workflows.in vision and representation learning, Anomaly detection.

Teaching Method: Lecture, in-class exercises

Evaluation:

Midterm Exam 30%

Final Exam 35%

Homework and assignments 35%

Grade	Score
A	≥ 85
B+	[84,80]
B	[79,75]
C+	[74,65]
C	[64,55]
D+	[54,50]
D	[49,45]
F	< 45

The instructor reserves the right to change the grading policy as deemed appropriate.

Key references:

- G. James, et.al., Introduction to Statistical Learning with Application with Python, 2023
- A. Geron, Hands-On Machine Learning with Scikit-Learn & Tensorflow, O'Reilly, 2019

Class policies:

1. Assignments are due in one week by submitting to LEB2.
2. Late submissions are only accepted under reasonable excuses and explicit permission from the instructor. No submission is accepted after the solution has been posted.
3. Posted solutions will be brief and do not show routine work. You should attempt to work out detailed solutions on your own.
4. Academic integrity is strictly enforced.